



**Caringbah High School**

**2015**

**Trial HSC Examination**

# Mathematics

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using a blue or black pen.  
Black pen is preferred.
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of the paper
- In Questions 11 – 16, show all relevant mathematical reasoning and/or calculations.

## Total Marks – 100

### Section I – 10 marks

- Attempt Questions 1 – 10
- Allow approximately 15 minutes for this section

### Section II – 90 marks

- Attempt Questions 11 – 16
- Allow approximately 2 hours and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

For Questions 1–10, use the multiple-choice answer sheet on page 18. Please detach this from the exam paper and submit with your answer booklets.

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- 1 Evaluate  $\log_e(\sin 2.5)$ , correct to 3 significant figures. 1

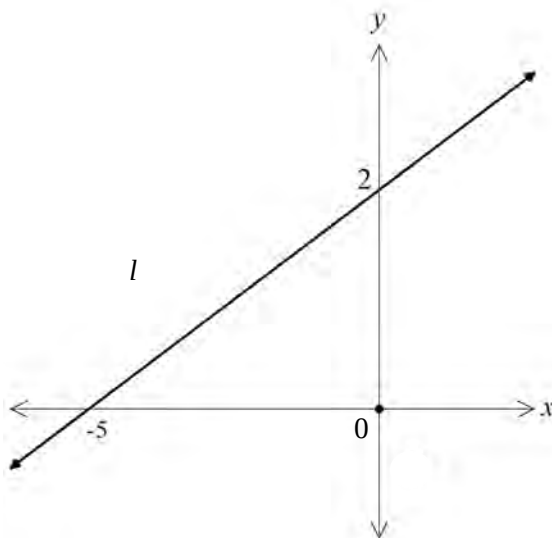
- (A)  $-0.223$
- (B)  $-0.513$
- (C)  $-1.36$
- (D)  $-3.13$

- 2  $(x + 2)$  is a factor of which expression. 1

- (A)  $x^3 - 8$
- (B)  $x^3 + 8$
- (C)  $x^2 + 4$
- (D)  $x^2 - 4x + 4$

- 3 The diagram shows the line,  $l$ . Find the gradient of the line,  $l$ . 1

- (A)  $\frac{-5}{2}$
- (B)  $\frac{-2}{5}$
- (C)  $\frac{2}{5}$
- (D)  $\frac{5}{2}$



4 What is the derivative of  $y = \frac{e^x}{e^x + 1}$  1

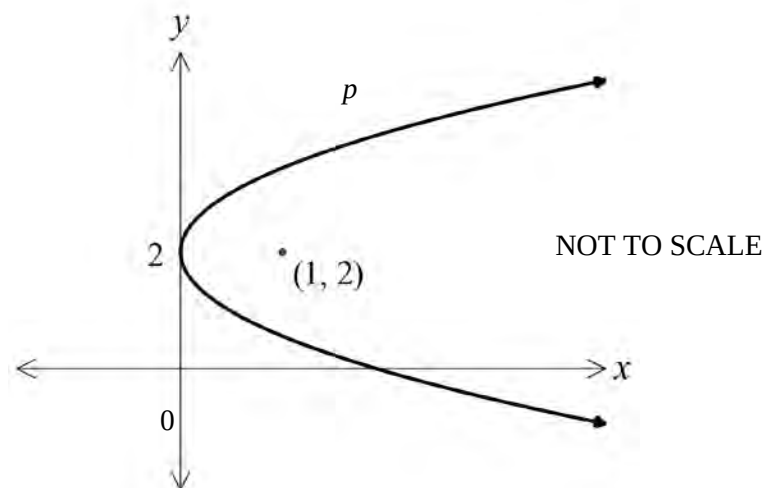
(A)  $\ln(e^x + 1)$

(B)  $\ln(e^x + 1)^2$

(C)  $\frac{2e^{2x} + e^x}{(e^x + 1)^2}$

(D)  $\frac{e^x}{(e^x + 1)^2}$

5 This graph shows the parabola,  $p$ , with vertex at 2 on the  $y$ -axis and focus  $(1, 2)$  1



Which equation represents the parabola,  $p$ ?

(A)  $y^2 = 8(x - 2)$

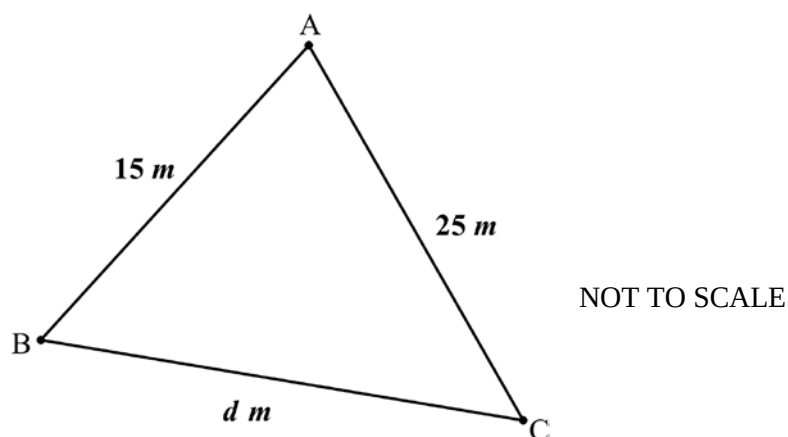
(B)  $(y - 2)^2 = 8x$

(C)  $y^2 = 4(x - 2)$

(D)  $(y - 2)^2 = 4x$

- 6 In  $\triangle ABC$   $AB = 15\text{ m}$ ,  $AC = 25\text{ m}$  and  $\angle ACB = 40^\circ$ .

1



Which expression can be used to find  $d$ ?

- (A)  $d^2 = 15^2 + 25^2 - 2 \times 15 \times 25 \cos 40^\circ$
- (B)  $d^2 = 15^2 + 25^2 - 2 \times 15 \times 25 \sin 40^\circ$
- (C)  $15^2 = 25^2 + d^2 - 2 \times d \times 25 \cos 40^\circ$
- (D)  $15^2 = 25^2 + d^2 - 2 \times d \times 25 \sin 40^\circ$
- 7 Find the primitive of  $4\sin(2x+3)$

1

- (A)  $-2\cos(2x+3)$
- (B)  $-8\cos(2x+3)$
- (C)  $2\cos(2x+3)$
- (D)  $8\cos(2x+3)$

8 Which inequality defines the domain of the function  $f(x) = \frac{x^2 - 9}{\sqrt{4 - x^2}}$  1

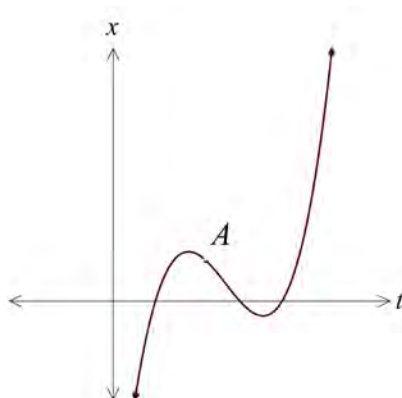
- (A)  $-2 < x < 2$
- (B)  $-3 < x < 3$
- (C)  $-2 \leq x \leq 2$
- (D)  $-3 \leq x \leq 3$

9 Which of the following describes the sequence 1

$$\ln(x) + \ln(x^2) + \ln(x^3) + \ln(x^4) + \dots$$

- (A) Arithmetic sequence
- (B) Geometric sequence with a limiting sum
- (C) Geometric sequence without a limiting sum
- (D) Neither an arithmetic nor a geometric sequence

10 The diagram shows the displacement,  $x$  metres, of a moving object at time  $t$  seconds. 1



Which of the following statements describes the motion of the object at the point A

- (A) Velocity is negative and acceleration is positive.
- (B) Velocity is negative and acceleration is negative.
- (C) Velocity is positive and acceleration is negative.
- (D) Velocity is positive and acceleration is positive

## Section II

**90 marks**

**Attempt Questions 11–16**

**Allow about 2 hours and 45 minutes for this section**

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use the Question 11 Writing Booklet.

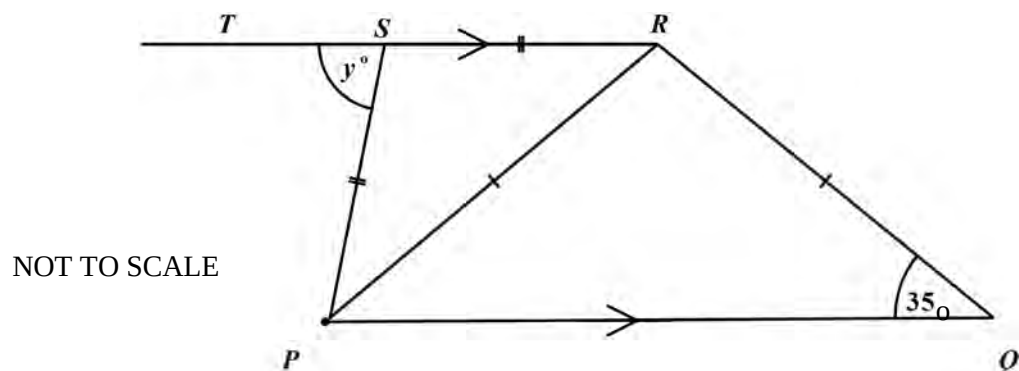
- (a) Given that  $x^5 = 10\,000$ , find  $x$  correct to 3 significant figures. 2
- (b) Evaluate  $\lim_{x \rightarrow 3} \frac{x^2 - x - 6}{x^2 - 9}$ . 2
- (c) Sketch the graph of  $y = x^3 - 8$ , showing intercepts with the coordinate axes. 2
- (d) Differentiate  $\log_e(\cos^2 x)$ , expressing your answer in simplest form. 2
- (e) Differentiate  $x^2 \sin x$ . 2
- (f) The derivative function of a curve  $y = f(x)$  is given by  $f'(x) = 6x^2 - 2$ . 2  
The curve passes through the point  $(1, 1)$ .

Find the equation of the curve.

**Question 11 continues on page 7**

Question 11 (continued)

- (g) The diagram shows a quadrilateral  $PQRS$ , in which  $PQ \parallel SR$ ,  $PS = SR$  and  $PR = RQ$ .  $T$  is a point on  $RS$  produced.



Copy or trace this diagram into your writing booklet.

- (i) Given that  $\angle RQP = 35^\circ$ , explain why  $\angle PRQ = 110^\circ$  1
- (ii) If  $\angle TSP = y^\circ$  find a value for  $y$  giving reasons. 2

**End of Question 11**

**Question 12** (15 marks) Use the Question 12 Writing Booklet.

(a)  $\int (2x-5)^3 dx$  2

(b) Evaluate  $\int_{\frac{\pi}{12}}^{\frac{\pi}{9}} \cos 3x \, dx$  leaving your answer in exact form. 3

(c) Evaluate  $\sum_{k=15}^{36} (3k-2)$ . 2

(d) The vertices of  $\triangle AOB$  are the points  $A(0, 4)$ ,  $O(0, 0)$  and  $B(6, -2)$ .

(i) Draw a diagram of  $\triangle AOB$  ( $\frac{1}{3}$  page), labelling all points clearly.

Also

mark in the point  $K$  that lies on the interval  $AB$  such that  $OK \perp AB$ .  
(You do not need to find the coordinates of  $K$ ).

1

(ii) By finding the gradient of  $AB$ , show that the equation of the line  $AB$  is given by,  $x + y - 4 = 0$ . 2

(iii) By finding the distance  $OK$ , show that the area of  $\triangle AOB$  is  $12u^2$ . 2

(iv) A horizontal line through  $B$  meets  $KO$  produced at  $S$ .  
Find the coordinates of  $S$ . 1

(v) Verify that  $AS \perp BO$ . 2

**End of Question 12**

**Question 13** (15 marks) Use the Question 13 Writing Booklet.

- (a) Sydney is 694 nautical miles (M) south of Noumea and 761 M west. 3  
Calculate the distance (to the nearest nautical mile) and bearing (to the nearest degree) a plane must fly to make a trip from Noumea to Sydney.

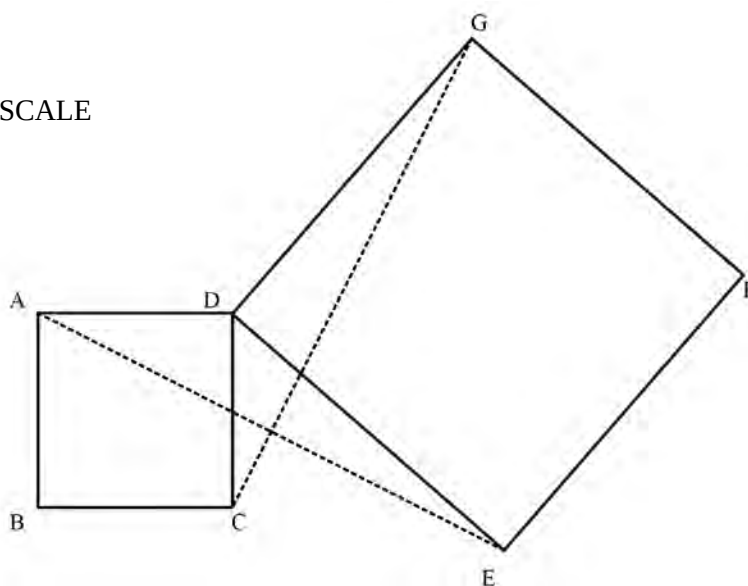
- (b) Sketch the graph of  $y = 3\sin 2x$  for  $0 \leq x \leq \pi$ . 2

- (c) If  $\sin x = \frac{-2}{5}$  and  $\cos x > 0$ , find the exact value of  $\cot x$ . 1

- (d) In this diagram  $ABCD$  and  $DEFG$  are both squares. 3

Prove  $AE = CG$ .

NOT TO SCALE

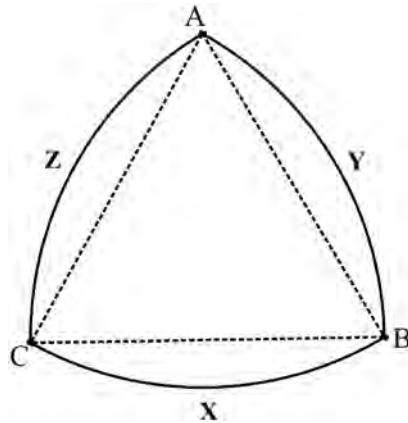


- (e) Find the equation of the normal to the curve  $y = \ln(3x + 4)$  at the point  $x = -1$  3

**Question 13 continues on page 10**

Question 13 (continued)

- (f) A cam in a lock,  $AYBXCZ$ , is drawn below.



NOT TO SCALE

$ABC$  is an equilateral triangle of side length 12cm. The arcs  $AYB$ ,  $BXC$  and  $CZA$  have centres  $C$ ,  $A$  and  $B$  respectively.

- |      |                                 |   |
|------|---------------------------------|---|
| (i)  | Find the perimeter of the cam.  | 1 |
| (ii) | Find the exact area of the cam. | 2 |

**End of Question 13**

**Question 14** (15 marks) Use the Question 14 Writing Booklet.

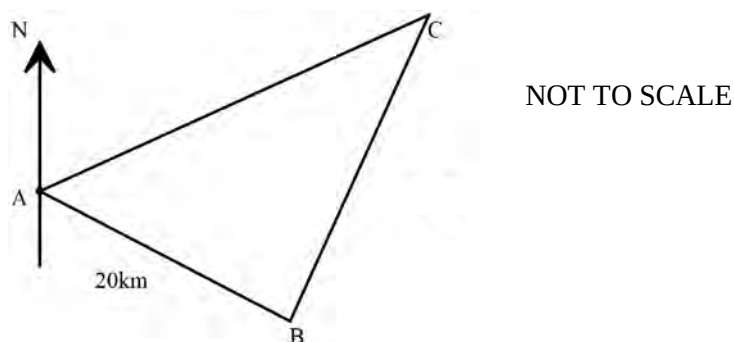
(a) Find the values of  $x$  for which  $|2x-1| \leq 3$ . 2

(b) Solve  $2 \sin\left(x - \frac{\pi}{4}\right) + 1 = 0$  for  $0 \leq x \leq 2\pi$ . 2  
Leaving your answer(s) in exact form.

(c) Find the values of  $m$  for which the graph of the parabola  $y = 4x^2 - mx + 9$   
(i) touches the  $x$ -axis. 1

(ii) crosses the  $x$ -axis. 1

(d) An ocean sailing regatta has yachts sailing in a triangular course as shown. 3



They sail from A for 20 km on a bearing of  $150^\circ$  T to the turning mark a B. The second leg from B to C is on a bearing of  $020^\circ$  T.

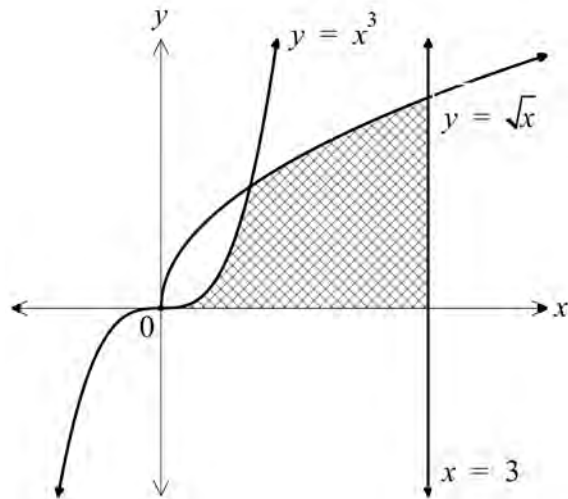
By determining the size of angle ABC, find the bearing of the third leg from C to A if it is known that C is 40 km from A.

**Question 14 continues on page 12**

Question 14 (continued)

- (e) The area bounded by the curves  $y = x^3$ ,  $y = \sqrt{x}$ , the  $x$ -axis and the line  $x = 3$  is shown.

3



NOT TO SCALE

Find the area of the enclosed region.

- (f) The area bounded by the curve  $y = \sec x$  and the  $x$ -axis for  $0 \leq x \leq \frac{\pi}{3}$  is rotated about the  $x$ -axis.

3

Find the volume of the solid of revolution formed by this rotation.

**End of Question 14**

**Question 15** (15 marks) Use the Question 15 Writing Booklet.

- (a) Solve these simultaneous equations. 2

$$4x - y = 3$$

$$10x + 3y = 2$$

- (b) A point  $P(x, y)$ , moves so that its distance from the line  $y = -2$  is equal to its distance from the point  $S(5, 2)$ . 2

Find the equation of the locus of  $P$ .

- (c) An object moves so that its velocity,  $v$  m/s, at any time,  $t$  seconds, is given by,

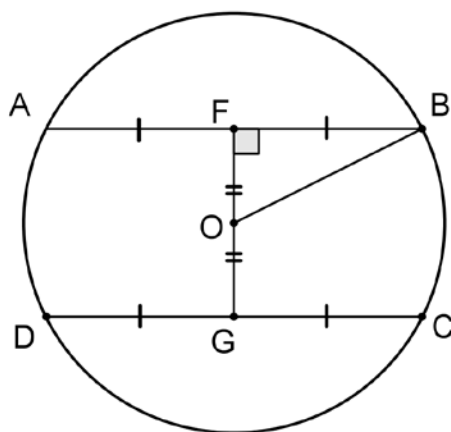
$$v = e^{-2t}$$

- (i) Show that the acceleration is always negative. 1
- (ii) Find the acceleration after 1 second. 1
- (iii) If the object is initially 2 m to the right of the origin, find an expression for the displacement of  $x$  in terms of  $t$ . 2
- (iv) Describe the motion of the particle as time increases. 2  
Include a description of displacement, velocity and acceleration.

**Question 15 continues on page 14**

Question 15 (continued)

- (d) A circular stained glass window of radius 3 m requires metal strips for support along  $AB$ ,  $DC$  and  $FG$ , as shown in the diagram.



Copy the diagram and information into your writing booklet.

$O$  is the centre of the circle.

Let  $OF = OG = y$  metres and  $FB = FA = GC = GD = x$  metres.

- (i) Find an expression for  $y$  in terms of  $x$ . 1
- (ii) The total length of the support strips (ie.  $AB + DC + FG$ ) is  $L$  metres. 1
- Show  $L = 4x + 2\sqrt{9 - x^2}$
- (iii) The window will have a maximum strength when the length of its supports is a maximum. 3

Show that  $FB = \frac{6\sqrt{5}}{5}$  metres, provides maximum strength for this window.

**End of Question 15**

**Question 16** (15 marks) Use the Question 16 Writing Booklet.

- (a) In January 2010 Jane decided to start saving for a holiday when she finishes Year 12.

She put \$10 into a shoe box for safe keeping. The next month, she added to her savings by placing \$12 in the shoe box. Each month she put in \$2 more than the previous month. In October 2015, Jane will place her final amount into the box.

- (i) How much will Jane put in the shoe box in October 2015? 2
- (ii) Calculate the total amount of her savings. 1  
(Note: Jane's shoe box does not offer any interest).

- (b) Radioactive substances decay over time. It is known that a mass,  $M$ , of a substance that remains after  $t$  years satisfies the equation,

$$M = M_0 e^{-kt}$$

Where  $M_0$  and  $k$  are constant.

The radioactive isotope radium 226 has a half-life of approximately 1600 years.

(Note: The half-life of a substance is the time taken for half the material to decay).

- (i) Find a value for  $k$ , expressing your answer in scientific notation correct to 3 significant figures. 1
- (ii) If ANSTO has a piece of radium 226 that weighs 200g, how much of this piece of radium existed in the year 1000 BCE (3015 years ago)? 2

**Question 16 continues on page 16**

Question 16 (continued)

- (c) The distance travelled by an object can be calculated by finding the area bounded by the graph of the velocity and the horizontal axis for time. 3

An object moves in a way that satisfies the equation,

$$v = 4 - \sin^2 t$$

Use Simpson's rule with 3 function values to find an approximate distance travelled by the object in the first 4 seconds.

- (d) At the completion of his degree, Oliver had a HEC's debt of \$100 000. He plans to repay this in equal monthly repayments of \$ $M$ . Interest is charged at a rate of 0.5% per month.

- (i) Show that the amount owing after 3 months,  $A_3$ , is given by 1

$$A_3 = 100\,000 \times 1.005^3 - M(1 + 1.005 + 1.005^2)$$

- (ii) If Oliver wishes to pay off his loan by the end of 10 years, then show that he will need to pay \$1 110 per month. 2

- (iii) Show that the amount owing after  $n$  months can be written as, 1

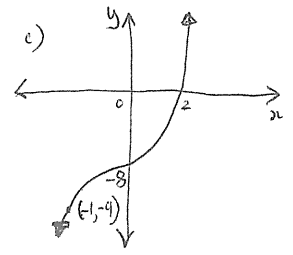
$$A_n = 1.005^n [100\,000 - 200M] + 200M$$

- (iv) If Oliver decides that he can only repay \$750 each month, how long will it take him to repay the loan? 2  
(Answer in years and months)

**End of paper**

M.C. B, B, C, D, D, C, A, A, B, B.

Q 11 a)  $\frac{6 \cdot 3}{(3 \cdot 5) \cdot (3 \cdot 5)}$   
 b)  $\lim_{x \rightarrow 3} \frac{(x-3)(x+2)}{(x-3)(x+3)} = \frac{5}{6}$



d)  $y = \ln(\cos^2 x)$   
 $y' = \frac{-2 \cos x \sin x}{\cos^2 x}$   
 $= -2 \tan x$

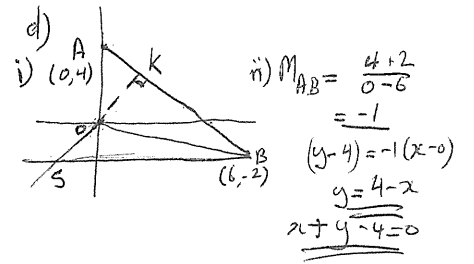
e)  $y = x^2 \sin x$   
 $y' = 2x \cos x + 2x \sin x$

f)  $y' = 6x^2 - 2$   
 $y = 2x^3 - 2x + c$   
 $x=1, y=1 \therefore c=1$   
 $y = 2x^3 - 2x + 1$

g) i)  $\angle RPQ = 35^\circ$  angles opp. eq sides of isos.  $\Delta$   
 $\angle PRQ = 110^\circ$  ( $180 - (35+35)$ ) angle sum  $\Delta$  RPQ  
 ii)  $\angle SRQ = 145^\circ$  co int  $\angle$ s,  $SR \parallel PQ$ .  
 $\therefore \angle SRP = 35^\circ$  ( $145 - 110$ )  
 $\angle SPR = 35^\circ$  angles opp eq. sides of isos  $\Delta$ .  
 $\therefore \angle TSP = 70^\circ$  ext  $\angle$  of  $\Delta$  SRP.

Q 12 a)  $\int (2x-5)^3 dx = \frac{2x-5^4}{8} + c$   
 b)  $\frac{1}{3} \left[ \sin 3x \right]_{\frac{\pi}{12}}^{\frac{\pi}{4}} = \frac{1}{3} \left[ \sin \frac{\pi}{3} - \sin \frac{\pi}{4} \right]$   
 $= \frac{1}{3} \left( \frac{\sqrt{3}}{2} - \frac{1}{\sqrt{2}} \right)$   
 $= \frac{\sqrt{3} - \sqrt{2}}{6}$

c)  $43 + 46 + 49 + \dots + 106$   $a=43, d=3$   
 $S_{22} = \frac{22}{2} [43 + 106]$   
 $= 1639$

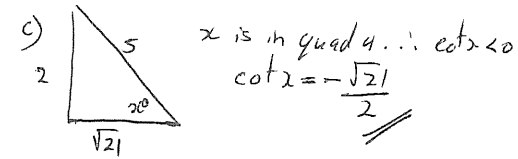
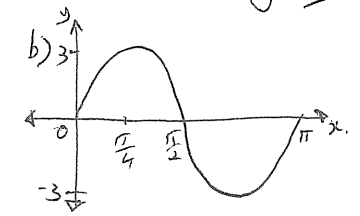
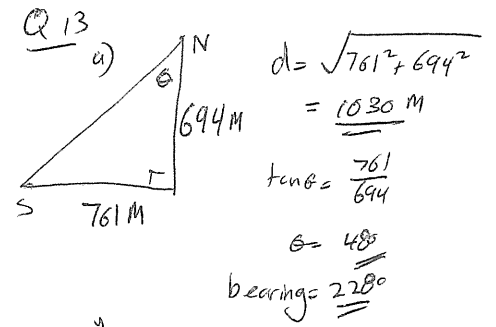


iii)  $OK = \frac{|0+0-4|}{\sqrt{1+1}}$   
 $= \frac{4}{\sqrt{2}}$

$A = \frac{1}{2} \times \frac{4}{\sqrt{2}} \times \sqrt{(4+2)^2 + (0-6)^2}$   
 $= \frac{1}{2} \times \frac{4}{\sqrt{2}} \times 6\sqrt{2}$   
 $= 12\sqrt{2}$

iv) Let S be  $(x, y)$   
 $m_{AS} = -1$   $\frac{y-0}{x-0} = -1$   $y = -x$   
 but  $y = -2$  so  $x = 2$   
 $S(2, -2)$

v)  $m_{AS} = \frac{6}{2}$   $m_{BO} = -\frac{2}{6}$   
 $m_{AS} \times m_{BO} = \frac{6}{2} \times -\frac{2}{6}$   
 $= -1$   
 $\therefore AS \perp BO$



d) In  $\Delta ADE$  and  $\Delta CDE$   
 $AD = CD$  and  $DE = CE$  sides of squares  
 If  $\angle DE = 20^\circ$   $\angle ADE = \angle CDE = 90 + 20$  ( $90^\circ$  in square)  
 $\therefore \Delta ADE \cong \Delta CDE$  SAS  
 $\therefore AE = CE$  corr. sides of cong  $\Delta$ s

e)  $y' = \frac{3}{3x+1}$   $x=1$   $m_T = 3$   $m_N = -\frac{1}{3}$   
 $x=1, y = \ln 1 = 0$   
 Eqn  $y-0 = -\frac{1}{3}(x-1)$   
 $x + 3y + 1 = 0$

f) i)  $L = r\theta$   $P = 3L$   
 $= 12 \times \frac{\pi}{3}$   $= 12\pi$   
 ii)  $A = \frac{1}{2} r^2 \theta + 2 \left( \frac{1}{2} r^2 (\theta - \sin \theta) \right)$   
 $= \frac{1}{2} \times 12^2 \times \frac{\pi}{3} + 2 \left( \frac{1}{2} \times 12^2 \left( \frac{\pi}{3} - \sin \frac{\pi}{3} \right) \right)$   
 $= 24\pi + \frac{144\pi}{3} - 144 \times \frac{\sqrt{3}}{2}$   
 $= 72\pi - 72\sqrt{3}$

Q 14 a)  $2x-1 \leq 3$   $2x-1 > -3$   
 $x \leq 2$   $x > -1$   
 $-1 \leq x \leq 2$   
 b)  $\sin(2 - \frac{\pi}{4}) = -\frac{1}{2}$   
 $x - \frac{\pi}{4} = \frac{7\pi}{6}$   $x - \frac{\pi}{4} = \frac{11\pi}{6}$   
 $x = \frac{17\pi}{12}$   $x = \frac{23\pi}{12}$   
 c)  $b^2 - 4ac = m^2 - 4 \times 36$   
 i)  $\Delta = 0$   $m = \pm 12$   
 ii)  $\Delta > 0$   $m < -12$  or  $m > 12$   
 d) 
 i)  $\angle ABC = 50^\circ$   
 $\frac{\sin \theta}{20} = \frac{\sin 50}{40}$   
 $\theta = 23^\circ$   
 $\therefore \text{bearing} = 223^\circ$

e)  $x^3 = \sqrt{x}$   $x = 0$  and  $x = 1$   
 $A = \int_0^1 x^3 dx + \int_1^3 x^{\frac{1}{2}} dx$   
 $= \frac{1}{4} [x^4]_0^1 + \frac{2}{3} [x^{\frac{3}{2}}]_1^3$   
 $= \frac{1}{4} + \frac{2}{3} (\sqrt{27} - 1)$   
 $= \frac{8\sqrt{27} - 5}{12}$

f)  $V = \pi \int_0^{\frac{\pi}{3}} \sec^2 x dx$   
 $= \pi [\tan x]_0^{\frac{\pi}{3}}$   
 $= \pi [\tan \frac{\pi}{3} - \tan 0]$   
 $= \sqrt{3}\pi$

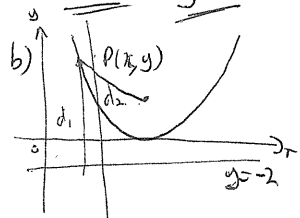
Q15 a)  $4x - y = 3$

$10x + 3y = 2$  (2)

$12x - 3y = 9$  (3)

(2)+(3)  $22x = 11$

$x = \frac{1}{2}$   $y = -1$



Parabola  $V(5, 0)$   $a = 2$ .

$(x-5)^2 = 8y$

$d_1^2 = d_2^2$   $\frac{0.2}{(y+2)^2} = \frac{(x-5)^2}{(y-2)^2}$   
 $(x-5)^2 = (y^2 + 4y + 4) - (y^2 - 4y + 4)$   
 $= 8y$

c) i)  $v = e^{-2t}$

$\frac{dv}{dt} = -2e^{-2t}$  as  $e^{-2t} > 0$

ii)  $t = 1$   $\frac{dv}{dt} < 0$ .

$a = -2e^{-2}$

iii)  $x = -\frac{1}{2}e^{-2t} + c$   
 $t = 0$   $x = 2$   $\therefore c = 2\frac{1}{2}$   
 $x = -\frac{1}{2}e^{-2t} + 2\frac{1}{2}$

iv) As  $t \rightarrow \infty$   $e^{-2t} \rightarrow 0$

$\therefore x \rightarrow 2\frac{1}{2}$

$v \rightarrow 0$ ,  $a \rightarrow 0$ .

ie slowing down approaching  $2\frac{1}{2}$ .

d) i)  $x^2 + y^2 = 9$   $\therefore y = \sqrt{9-x^2}$

ii)  $L = 4x + 2y = 4x + 2\sqrt{9-x^2}$

iii)  $L' = 4 + (9-x^2)^{-\frac{1}{2}}x - 2x$

$\max L' = 0$   $\frac{2x}{\sqrt{9-x^2}} = 4$

$\frac{4x^2}{9-x^2} = 16$

$4x^2 = 144 - 16x^2$

$20x^2 = 144$

$x = \frac{12}{\sqrt{20}}$  (as  $x > 0$ )

$= \frac{6\sqrt{5}}{5}$

check for max

$x \left| \begin{array}{l} 1 \\ < 0 \end{array} \right| \frac{6\sqrt{5}}{5} \left| \begin{array}{l} 2.9 \\ > 0 \end{array} \right|$

$\therefore \max$  at  $x = \frac{6\sqrt{5}}{5}$

Q16

a)  $\$10 + \$12 + \$14 + \dots$

$a = 10$   $d = 2$   $n = 5 \times 12 + 10$   
 $= 70$

i)  $T_n = 10 + 69 \times 2$

$= \$148$

ii)  $S_n = \frac{70}{2} (10 + 148)$

$= \$5530$

b) When  $t = 1600$   $M = \frac{1}{2}M_0$

i)  $\frac{1}{2}M_0 = M_0 e^{-1600k}$

$-1600k = \ln \frac{1}{2}$

$k = -\frac{\ln \frac{1}{2}}{1600}$   $\left( \frac{\ln 2}{1600} \right)$

$= 4.33 \times 10^{-4}$

ii)  $200 = M_0 e^{-3015k}$

$M_0 = 200 \times e^{3015k}$

$= 738g$

c)  $d = \int_0^4 4 - \sin^2 t$

|   |                |                |
|---|----------------|----------------|
| 0 | 2              | 4              |
| 4 | $4 - \sin^2 2$ | $4 - \sin^2 4$ |

$R = 2$

$d \approx \frac{2}{3} [4 + (4 - \sin^2 2) + 4(4 - \sin^2 4)]$

$= 13.4m$

Q16d

i)  $A_1 = 100000 \times 1.005 - M$

$A_2 = 100000 \times 1.005^2 - M \times 1.005 - M$

$A_3 = 100000 \times 1.005^3 - M \times 1.005^2 - M \times 1.005 - M$   
 $= 100000 \times 1.005^3 - M(1 + 1.005 + 1.005^2)$

ii) 10 yrs  $\Rightarrow n = 120$

$A_n = 100000 \times 1.005^n - M(1 + 1.005 + \dots + 1.005^{n-1})$

$M = \frac{100000 \times 1.005^{120}}{\left[ \frac{1.005^{120} - 1}{0.005} \right]}$

$= \$1110.20$

iii)  $A_n = 100000 \times 1.005^n - M \left[ \frac{1.005^n - 1}{0.005} \right]$

$= 100000 \times 1.005^n - 200M [1.005^n - 1]$

$= 1.005^n [100000 - 200M] + 200M$

iv)  $1.005^n [100000 - 200 \times \$750] + 200 \times \$750 = 0$

$1.005^n = -\frac{200 \times 750}{100000 - 200 \times 750}$   
 $= \frac{150000}{50000}$

$n = \frac{\ln 3}{\ln 1.005}$

$= 220 \text{ mths}$

$= 18 \text{ yrs } 5 \text{ mths}$